

BRIAN LANE

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SENIOR SOFTWARE ENGINEER with a track record of building scalable, high-performance backend systems across finance and aerospace. Experienced in microservices, cloud solutions, and backend architecture, with a strong focus on practical, efficient problem-solving through code.

AREAS OF FOCUS

Microservices Architecture | Cloud Computing (AWS Certified) | Agile & Scrum Methodologies | Systems Programming

Tech: Python | Java | Spring Boot | SQL | Docker | Kotlin | GitHub | pgAdmin | Splunk | New Relic | Scripting | Jenkins

PROFESSIONAL EXPERIENCE

American Express | July 2025-Present

Senior Software Engineer – Corporate Card Management

Engineered API endpoints and company wide authorization security.

- Manufactured end to end software using Kotlin including architecture diagrams, reliable production code, and thoughtful unit tests for dozens of new API endpoints using Ktor standard RESTful practices.
- Documented workflow process for end to end work of hundreds of REST API endpoints to increase efficiency from more than 2 weeks to 2 days.
- Refactored old code to meet new company wide authorization security standards for token generation for dozens of REST API endpoints.
- Provided valuable knowledge transfers to team members and new onboarding members.

Capital One | June 2022-April 2025

Senior Software Engineer – Prism Team

Collaborated on Java Spring boot backend development and system optimization efforts.

- Increased operational efficiency by improving data collection processes for allowlists and filter data, resulting in the automation of a full day of manual work.
- Decommissioned legacy web and mobile apps, documenting migration and prepping new teams to move thousands of REST API endpoints and filters without data loss during live updates to new microservices.
- Developed a ReactJS self-service portal for hundreds of internal clients using backend web technologies, reducing support requests.
- Achieved AWS Certified Solutions Architect status, deepening cloud development capabilities and contributing to cloud-first engineering initiatives.

Charles Schwab | June 2019-June 2022

Software Engineer – Human Resource Technology Team

Contributed to the design and delivery of internal HR applications, including scalable backend systems.

- Built and deployed Java Spring Boot microservices on Pivotal Cloud Foundry with CI/CD pipelines via Bamboo, improving release speed and service reliability.
- Developed HR data exchange for Workday traffic by collaborating on Java Spring Boot microservices architecture, enabling Java applications to support both full-time and part-time employees.
- Created Java Spring Boot architecture to seamlessly manage contractor data, resulting in the successful logging of hundreds of contractors and their information.
- Completed intensive new hire training program, earning Java and Scrum Master certifications to strengthen Agile delivery and technical execution.

Honeywell | August 2018-June 2019

Software Engineer – User Interface & Guidance Team

Supported the development and quality assurance of aerospace software systems

- Performed system-level testing on flight simulators, gaining hands-on aerospace bench experience and strengthening real-world application insight.
- Led code, requirements, and design reviews to enforce quality standards and deliver bug-free software aligned with mission-critical performance goals.
- Completed Six Sigma Green Belt Software Development and Design Training, equipping the team with process improvement strategies and defect reduction methods.

PERSONAL EXPERIENCE

Home Assistant

- Fully automated home with custom scripts and integrations using a Raspberry Pi
- Originally developed as a simple home automation system that evolved into a full-fledged smart home solution
- Implemented custom automations and preferred logging methods

Technologies: Raspberry Pi, Scripting, Yaml Configuration, Zwave/Zigbee

Surveillance Camera

- Developed a monitoring system using Raspberry Pi and Camera Module V2
- Conducted performance testing comparing OpenCV and Google Cloud Vision APIs for optimal recognition in live stream

Technologies: Raspberry Pi, OpenCV, Google Cloud Vision, Bash

caseFix | iOS Application

- Created an iOS application that enhances keyboard functionality on iOS devices
- Developed with Swift to modify text case (lowercase, uppercase, or fix case)
- Built solution particularly useful for voice-to-text processes

Technologies: iPhone, Swift, iOS, AppStore

BizBlasts | Ruby Monolithic App | www.bizblasts.com

- Made a comprehensive booking system to accept payments with a fully integrated e-commerce virtual storefront to allow for the functionality of services, products, rentals, and much more.
- Featured real-time inventory management and Stripe payment processing for small business, provided CRM for users

Technologies: Ruby, Render, PostgreSQL, GitHub Actions

New Coworker | ReactJS App | www.newcoworker.com

- Gave businesses the ability to answer calls, texts, emails, and run automation with dedicated AI, number and email
- Used lossless memory to learn about each business over time so every interaction builds context

Technologies: React, Javascript, PostgreSQL, GitHub Actions

View all my upcoming projects on my github -> <https://github.com/brianlane>

EDUCATION

Bachelor of Science in Computer Science, Minor in Psychology, University of Arizona, College of Science, Tucson, AZ

AWS Certified Solutions Architect – Associate, Capital One

Java & Scrum Master (PSM I) Certified, Charles Schwab

Certification in Six Sigma Green Belt Software Development & Design, Honeywell

Microsoft Technology Associate: Software Development Fundamentals (MTA), Microsoft